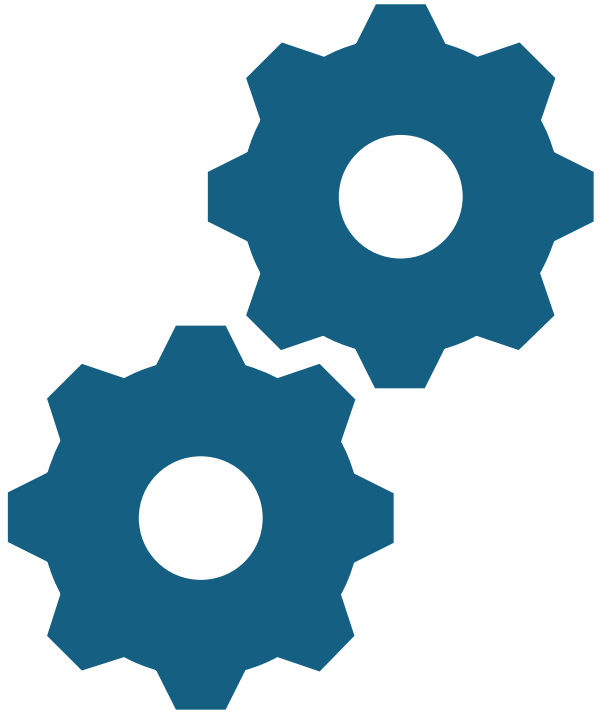
# Advanced Deterministic Systems Optimization



ISE 555 Unit 4

# Re-cap

- Last week we looked at the Standard Form for Linear Programs
- Examined how to convert into standard form and how to find extreme points efficiently
- Next up:  The Simplex Method for solving linear programs
- Note: This is in Chapter 3.10—3.11 if you are using the textbook for reference

# Review of Standard Form

- Standard form puts all linear programs into the following:

$$\begin{aligned} \text{minimize} \quad & z = c^T x \\ \text{subject to} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

$$\text{with} \quad b \geq 0$$

**3 MAIN PARTS**

**OBJECTIVE**

**REGULAR CONSTRAINTS**

**BOUNDS ON VARIABLES**

- NOTES:
  - Recall that LINEAR PROGRAM means both the objective and constraints are linear functions of the variables
  - OBJECTIVE must be (1) minimization and (2) not have any additive constants
  - REGULAR CONSTRAINTS must be (1) all equalities and (2) have all right-hand sides non-negative
  - BOUNDS ON VARIABLES must (1) exist for all variables (even those we added) and (2) all be of the form $\geq 0$

# Question: How to deal with different bounds on the variables?

- Example 1: $x_1 \geq 5$
  write as $x_1 = x' + 5 \quad \Rightarrow \quad x' \geq 0$

- Example 2: $x_2$ free
  write as $x_2 = x'_2 - x''_2 \Rightarrow \quad x'_2 \geq 0, \;\; x''_2 \geq 0$

- Example 3: $0 \leq x_3 \leq 4$
  write as $x_3 \geq 0$ and add $x_3 \leq 4$ to regular constraints $\Rightarrow \; x_3 + s_4 = 4, s_4 \geq 0$

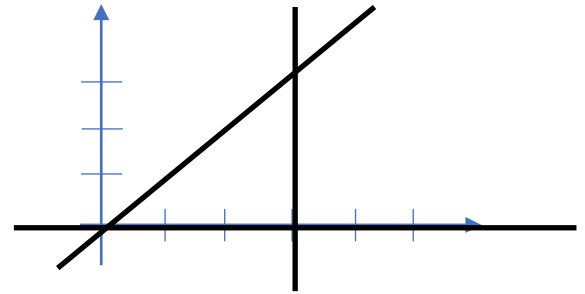
- Example 4: $x_4 \leq 6$
  (a) add $x_4 \leq 6$ to regular constraints and make $x_4$ free so $x_4 + s_4 = 6$
  $x_4 = x'_4 - x''_4, \;\; x'_4 \geq 0, \;\; x''_4 \geq 0, \;\; s_4 \geq 0$
  (b) notice that $-x_4 \geq -6$
  $\Rightarrow \; -x_4 + 6 \geq 0$ and let $x_4 = -x'_4 + 6 \; \Rightarrow \; x'_4 \geq 0$

# White Board Examples

# Example of standard form and feasible points

- Find the basic solutions and determine their feasibility for the following set of constraints:

$$x_1 - x_2 \geq 0$$
$$x_1 \leq 3$$
$$x_2 \geq 0$$



- VERSION 1:

$$-x_1' + x_1'' + x_2 + s_1 = 0$$
$$x_1' - x_1'' + s_2 = 3$$
$$x_1' \geq 0$$
$$x_1'' \geq 0$$
$$x_2 \geq 0$$
$$s_1 \geq 0$$
$$s_2 \geq 0$$

so $A = \begin{bmatrix} -1 & 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 0 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$

# Example of standard form and feasible points

a) Try $x_B = [x_1' \; x_2]^T$ so that $B = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$, $x_B = B^{-1}b = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$, $x_1 = x_1' - x_1'' = 3$ , $[x_1, x_2] = [3,3]$

b) Try $x_B = [x_1' \; s_1]^T$ so that $B = \begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$, $x_B = B^{-1}b = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$, $x_1 = x_1' - x_1'' = 3$, $[x_1, x_2] = [3,0]$

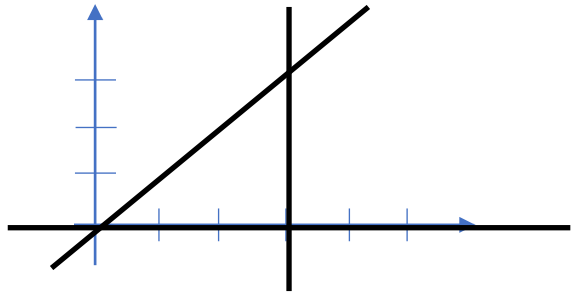
c) Try $x_B = [s_1 \; s_2]^T$ so that $B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $x_B = B^{-1}b = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$, $[x_1, x_2] = [0,0]$

d) Try $x_B = [x_1'' \; x_2]^T$ so that $B = \begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$, $B^{-1} = \begin{bmatrix} 0 & -1 \\ 1 & 1 \end{bmatrix}$, $x_B = B^{-1}b = \begin{bmatrix} -3 \\ 3 \end{bmatrix}$, $x_1 = x_1' - x_1'' = 3$, $[x_1, x_2] = [3,3]$

# Example of standard form and feasible points

- Find the basic solutions and determine their feasibility for the following set of constraints:

$$x_1 - x_2 \geq 0$$
$$x_1 \leq 3$$
$$x_2 \geq 0$$



- VERSION 2:

Let $x_1' = 3 - x_1 \quad \Rightarrow \quad x_1' \geq 0$

$$x_1' + x_2 + s_1 = 3$$
$$x_1' \geq 0$$
$$x_2 \geq 0$$
$$s_1 \geq 0$$

so $\quad A = [1 \quad 1 \quad 1]$ and $\quad b = [3]$

# Example of standard form and feasible points

a) Try $x_B = [x_1']$ so that $B = [1],\ B^{-1} = [1],\ x_B = B^{-1}b = [3],\ x_1 = 3 - x_1' = 0\ ,\ [x_1, x_2] = [0,0]$

b) Try $x_B = [x_2]$ so that $B = [1],\ B^{-1} = [1],\ x_B = B^{-1}b = [3],\ x_1 = 3 - x_1' = 3\ ,\ [x_1, x_2] = [3,3]$

c) Try $x_B = [s_1]$ so that $B = [1],\ B^{-1} = [1],\ x_B = B^{-1}b = [3],\ x_1 = 3 - x_1' = 3\ ,\ [x_1, x_2] = [3,0]$

# Getting from Basic Feasible Solutions to an Optimal Solution

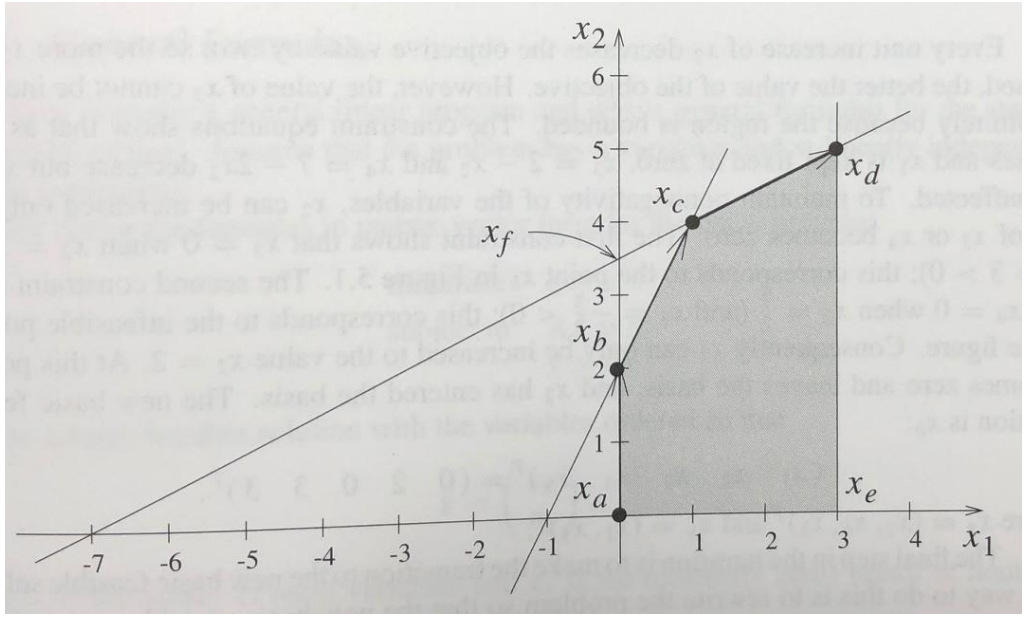
- Since any optimal solution must be a basic feasible solution, we can just do the following:
  1. Find the set of all basic feasible solutions
  2. Evaluate the objective at all basic feasible solutions
  3. Pick the basic feasible solution with lowest objective value as optimum

- Alternatively we can move through basic feasible solutions efficiently by using what we know about feasible directions:
  1. Start at one basic feasible solution
  2. Determine if there exists a feasible descent direction – if not STOP as it is the optimum
  3. Otherwise move to an adjacent basic feasible solution that follows a descent and repeat

- This alternative approach is **The Simplex Method**

# Example to show The Simplex Method approach

$$\begin{aligned} \text{minimize} \quad & z = -x_1 - 2x_2 \\ \text{subject to} \quad & -2x_1 + x_2 \leq 2 \\ & -x_1 + 2x_2 \leq 7 \\ & x_1 \leq 3 \\ & x_1, x_2 \geq 0 \end{aligned}$$




OPTIMUM

| POINT | LABEL | ACTIVE CONSTRAINTS | VIOLATED CONSTRAINTS | FEASIBLE OR INFEASIBLE | $Z$ |
|---|---|---|---|---|---|
| [0, 0] | $x_a$ | B1, B2 | None | Feasible | 0 |
| [0, 2] | $x_b$ | R1, B1 | None | Feasible | -4 |
| [1, 4] | $x_c$ | R1, R2 | None | Feasible | -9 |
| [3, 5] | $x_d$ | R2, R3 | None | Feasible | -13 |
| [3, 0] | $x_e$ | R3, B2 | None | Feasible | -3 |
| [0, 3.5] | $x_f$ | R2, B1 | R1 | Infeasible | N/A |
| [-1, 0] | | R1, B2 | B1 | Infeasible | N/A |
| [-7, 0] | | R2, B2 | B1 | Infeasible | N/A |
| [3, 8] | | R1, R3 | R2 | Infeasible | N/A |
| NONE (PARALLEL) | | R3, B1 | N/A | Infeasible | N/A |

# What is an Optimal Solution

- Assume we have a linear programming problem in *standard form* with $A$ of dimension $m \times n$ (and $m < n$)
- Recall that a *basic solution* is found by forming matrix B as a square subset of matrix A by choosing m columns of A
- Call the variables corresponding to these columns the *basic variables* $x_B$
- Now $x_B = B^{-1}b$ yields the values of the basic variables, the other variables are all zero to meet the constraints $Ax = b$
- If the basic variable values are all non-negative, the point with values $x_B$ is a *basic feasible solution* – corresponding to an extreme point (or corner)
- If we replace any basic variables in the objective with non-basic variables (using equalities from the constraints), then the objective is written in terms of variables that are zero at our basic feasible solution.
- If all coefficients at positive, then the solution cannot be improved (meaning lowering the objective) by increasing any of the non-basic variables. Thus, the basic feasible solution is optimal!

White Board Example

# The Simplex Method Algorithm

- Assume a problem in standard form where $Ax = b$ and goal is to minimize $z = c^T x$ under conditions of $x \geq 0$

The Simplex Method is performed in four basic steps:

1. **INITIALIZATION:**

   Find a basic feasible solution $x_B$ with basis matrix $B$ (such that $x_B = B^{-1}b$). Call the remaining variables $x_N$.

2. **OPTIMALITY TEST:**

   Write the objective in form $z = c_B^T x_B + c_N^T x_N$ and constraints in the form $Bx_B + Nx_N = b$. Compute coefficients $\hat{c}_N^T = c_N^T - c_B^T B^{-1} N$. If all components are non-negative – then it is optimal; otherwise pick one that is negative and the corresponding variable becomes the entering variable $x_t$.

3. **STEP:**

   Compute $\hat{A}_t = B^{-1} A_t$ as the constraints corresponding to the entering variable. Find an index $s$ corresponding to the leaving variable such that

$$\frac{\hat{b}_s}{\hat{a}_{s,t}} = \min_{1 \leq i \leq m} \left\{ \frac{\hat{b}_i}{\hat{a}_{i,t}} : \ \hat{a}_{i,t} > 0 \right\}$$

4. **UPDATE:**

   Update the basis $x_B$ by replacing the leaving variable with the entering variable and compute a new basis matrix $B$

# Some Issues

- Deficiencies in tableau approach
  - Does a lot more algebra than is required
  - Does not take advantage of sparsity of the problem (which is important in practical implementations)
  - Initial tableau is only obvious when all constraints start as $\leq$ style
- What happens for unbounded regions?

# Summary

- I will post a PDF version of these slides on BrightSpace
- Today we saw how to do the Simplex Method and how to use a tableau
- Next class we will continue on the Simplex Method and discuss Duality
- No class next week because of Indigenous Peoples Holiday – next meet on Oct 17
- HW #4 is due in two weeks (on Oct 19)